

Rectangle Cutting

Time limit: 1.00 s

Memory limit: 512 MB

Given an $a \times b$ rectangle, your task is to cut it into squares. On each move you can select a rectangle and cut it into two rectangles in such a way that all side lengths remain integers. What is the minimum possible number of moves?

Input

The only input line has two integers a and b .

Output

Print one integer: the minimum number of moves.

Constraints

- $1 \leq a, b \leq 500$

Example

Input	Output
3 5	3